

# JEE MAIN REVISION MASTER: ALCOHOL, PHENOLS AND ETHERS 2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                       | Frequency | Category        |
|---------------|----------------------------------------------------|-----------|-----------------|
| Very High     | Kcet Alcohol Phenols And Ethers - Core Application | 32 PYQs   | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Kcet Alcohol Phenols And Ethers - Core Application (Very High Yield — 32 PYQs)

**Core Principle:** Key problem pattern in Kcet Alcohol Phenols And Ethers.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

**Representative Source PYQs:**

- [2026 Shift] Q#1 (MCQ): "The correct IUPAC name of is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Glycerol is a trihydric alcohol. It contains \_\_\_\_." [Ans: B]
- [2026 Shift] Q#3 (MCQ): " $R-CH_2OH$  is converted into  $R-CHO$  by reacting with \_\_\_\_." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: KCET 2026 (Q#1)**

The correct IUPAC name of is

- (A) Tertiary butoxy ethane
- (B) 1,1-Dimethyl-1-ethoxyethane
- (C) 2-ethoxy-2-methyl propane
- (D) Ethoxy tertiary butane

**Correct Answer: None**

**Solution Analysis:** In an ether, the larger carbon part is chosen as the parent hydrocarbon, and the smaller part attached through oxygen is named as an alkoxy group. Here, one side of oxygen is an ethyl group,  $-C_2H_5$ . So this part becomes the alkoxy group: ethoxy,  $-OC_2H_5$ . The other side is a tert-butyl group,  $-C(CH_3)_3$ . The longest continuous chain in this part has 3 carbon atoms, so the parent hydrocarbon is propane. In the propane chain, the middle carbon is carbon number 2. Both substituents are attached to this carbon. At carbon 2, there is one methyl group,  $-CH_3$ , and one etho